Module 4

Shell Productivity & Basic Scripting

Learning objectives

Use history and Ctrl+R effectively

Write scripts with shebang and chmod +x

Pass arguments with $1 and $@

Apply set -e and quoted variables

Prerequisites

Module 3 (redirection, pipes)

Learning path

Learning Path

```
Diagram: Learning Path
(render with mmdc when available)
flowchart LR
A[History & aliases] --> B[Scripts]
B --> C[Variables]
C --> D[Control flow]
D --> E[Safe scripting]
E --> F[Module 5]
```

Course context

Wrap compile/sim steps in scripts/

Course module scripts are the pattern to follow

Quote variables to avoid word-splitting bugs

Core topics

Self-check

`$ ./scripts/module4.sh --check`

Terminal: module_check

```
□[0;36m========================================□[0m
□[0;36mModule 4: Environment self-check□[0m
□[0;36m========================================□[0m

□[0;32m[OK]□[0m bash available
□[0;32m[OK]□[0m module4 directory exists: /home/yongfu/proj/universal-verification-methodology/learn
□[0;32m[OK]□[0m script_basics/say_hello.sh exists
□[0;32m[OK]□[0m arguments/greet.sh exists
□[0;32m[OK]□[0m control_flow/count_lines.sh exists
□[0;32m[OK]□[0m safe_scripting/clean_tmp.sh exists

□[0;32m[OK]□[0m All required checks passed.
```

Minimal script

```
#!/usr/bin/env bash
echo "Hello from script"
```

First bash script

```
$ cd module4/examples/script_basics && chmod u+x say_hello.sh && ./say_hello.sh
```

Positional parameters

```
$ cd module4/examples/arguments && chmod u+x greet.sh && ./greet.sh Student
```

Terminal: greet_args

```
Hello, Student!
```

Practice

Exercise scaffold

```
./scripts/module4.sh --scaffold
```

Exercises

Run say_hello.sh and greet.sh from scaffold

Add an alias ll='ls -alF'

Write a loop over *.log

Summary & next steps

Scripts automate design flows

History and aliases save time

Next: Module 5 — projects & archives

Next: Module 5: Editors, Projects & Archives